



Exploring Undergraduate Students' Experiences with Standards-Based Grading

Russ deForest, Debbie Gaydos, Neil Hatfield, Joe Houck, Eric Hudson, Louis Leblond, Michael Steward

The Pennsylvania State University



Specifications or Standards-based grading (SBG) is a set of grading approaches that focus on evaluating students' proficiency of learning objectives. Often graded on pass/fail or proficient/progressing with multiple opportunities for retakes.

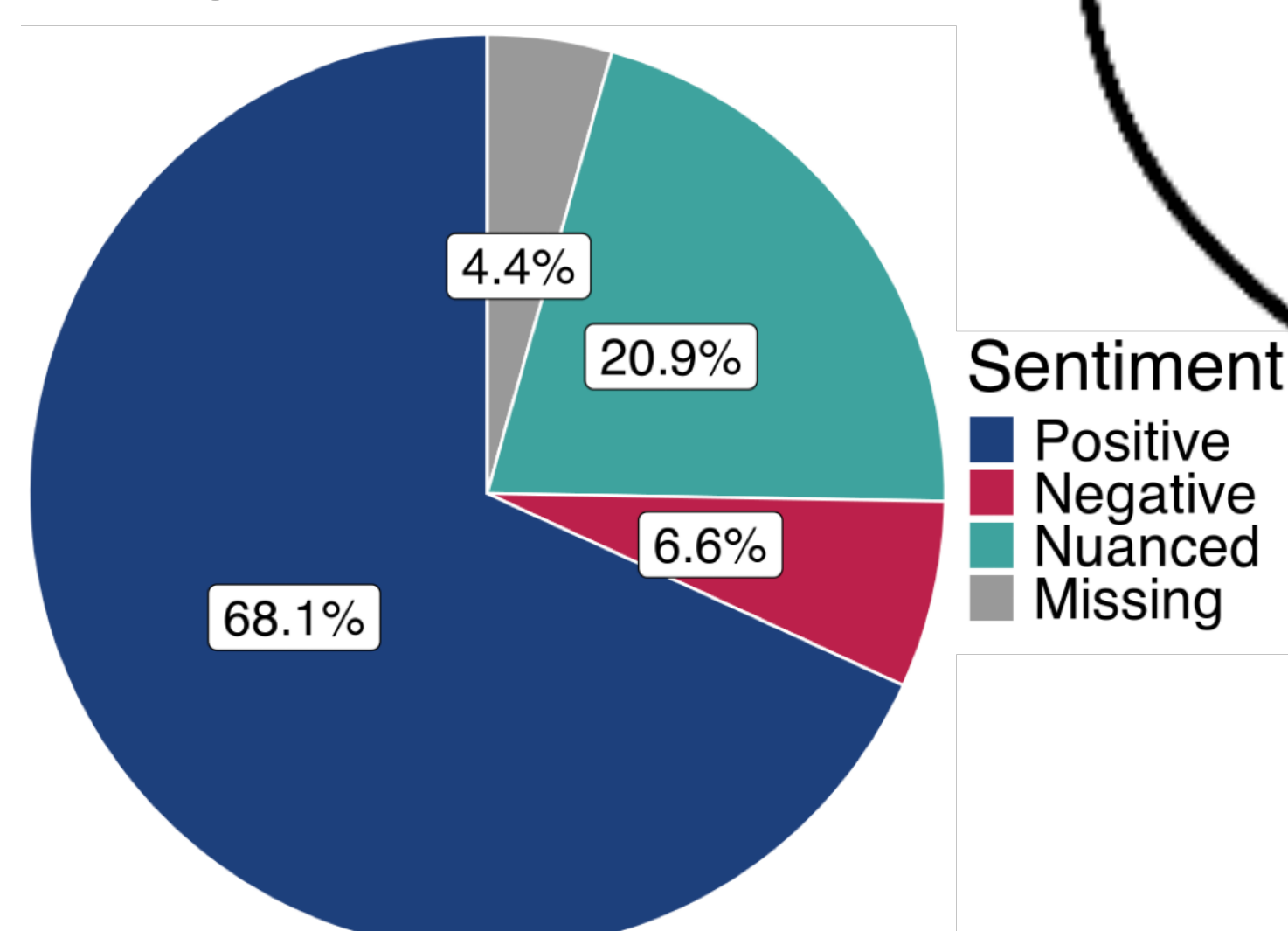
Research Question

What are the experiences of undergraduate students in science and mathematics courses that use Standards-Based/Specifications Grading?

Courses	N	Main Assessment
Intro Calculus	238	Learning Target Quizzes with retakes
Statistics	132	Homework/Projects
Chemistry	739	Learning Target Quizzes with retakes
Physics	90	Learning Target Quizzes with retakes

METHODS

1. Retrospective Survey data (91 respondents).
2. Individual and focus group interviews (24 students to date).
3. Focused coding, memos.



Manageable set of learning targets (LT)
(e.g., 20-30 for a 15-week course)

Satisfactory/progressing
No partial credits (e.g., weekly quizzes)

Bundled assignments for letter grade
(e.g., 20/25 proficient LT => B)

Our most common implementation

Retakes as needed

"Experience points"
(e.g., > 90 points on "readings" bumps to B+)

Students perceive that the control and flexibility of standards-based grading increases motivation and reduces stress but some report procrastination and test-reset behaviors.

[...] and just gather knowledge at your own pace, you know, at your own time, and stress-free.

You were able to just breathe and really understand the work that you were doing

You really had full control over what grade you were going to get.



Some of the Findings

SBG = Standards-Based grading
TG = Traditional Grading

Grades as source of motivation

- SBG: Low grades are not damning
- SBG: Grades only increase
- TG: Disengage if you feel your grade cannot change

Understanding material/content

- SBG/TG: Test/reset phenomenon
Learn to demonstrate performance and then forget.

Varying Views on the following

- SBG: Chunking and no partial credits help learning
- TG: Final cumulative exam helps learning

Stress of grading systems

- TG: Constant risk of losing a grade
- SBG: Unclear what your grade is till the end.

Control over learning

- SBG: Grade is a result of efforts
- TG: Disincentivizes learning from mistakes

References

- Nilson, L. B. Specifications Grading: Restoring Rigor, Motivating Students, and Saving Faculty Time. Stylus, 2014.
- Talbert, R.; Clark, D. Grading for Growth, www.gradingforgrowth.com

Acknowledgement:

We thank Jennelle Malcos for help with focus groups and the Eberly College of Science for a teaching innovation grant.